Reviewer Pack — Minimal Hodge Index of Quasi-Clifford Algebras and Sum-of-Sq...

Entry d8441229b8b9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Index of Quasi-Clifford Algebras and Sum-of-Squares Refutation Size

Entry ID: `d8441229b8b9` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-22 01:05:07 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory: Quasi-Clifford Algebras **Field B** (complexity object): Sum-of-Squares Hierarchy for CSP Lower Bounds

Statement:

> ['For every instance of k-CNF with n variables, the minimal Hodge index of the associated quasi-Clifford algebra is upper-bounded by the square root of the size of its smallest refutation in the Sum-of-Squares (SOS) hierarchy.', 'Equivalently, if an instance has a refutation of size s in the SOS hierarchy, then the minimal Hodge index of its corresponding quasi-Clifford algebra satisfies:', 'HodgeIndex(Quasi Clifford Algebra(CNF)) ≤ √s']

Rationale (proposer's reasoning):

> ['Quasi-Clifford algebras are known to have a rich geometric structure that could potentially encode non-trivial complexity information.', 'The Hodge index is an algebraic invariant that captures the complexity of certain geometric objects and might relate to the difficulty of refuting formulas in the SOS hierarchy.', 'This conjecture would provide a new avenue for understanding the interplay between representation theory and computational complexity.']

Taxonomy category: `SOS_HIERARCHY` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `2938f317ff494dd7`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The minimal Hodge index of a Quasi-Clifford Algebra for a given k-CNF instance is supported if it does not exceed the square root of its smallest SOS refutation size by an absolute difference of at most 3, across all seeds.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `"quasi-Clifford algebras" AND "Sum-of-Squares Hierarchy"` - `"minimal Hodge index" AND "SOS refutation size"` - `"CNF and quasi-Clifford algebra" AND "HodgeIndex ≤ √s"`

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_k_cnf(n, k):
        variables = list(range(1, n + 1))
        clauses = []
        for _ in range(k):
            clause = [random.choice(variables), random.choice(variables)]
            if random.choice([True, False]):
                clause[0] *= -1
            if random.choice([True, False]):
                clause[1] *= -1
            clauses.append(clause)
        return clauses

    def hodge_index(n):
        # Placeholder for actual Hodge index calculation
        # This is a dummy implementation that returns a simple function of n
        return n * (n + 1) // 2

    def sos_refutation_size(k_cnf):
        # Placeholder for actual SOS refutation size calculation
        # This is a dummy implementation that returns a simple function of k_cnf
        return len(k_cnf) ** 2

    n = random.randint(5, 40)
    k = random.randint(1, n * (n - 1) // 2)
```

```python
    k_cnf = generate_k_cnf(n, k)

    hodge_val = hodge_index(n)
    sos_size = sos_refutation_size(k_cnf)

    if sos_size == 0:
        return {
            "metric_name": "Hodge Index",
            "metric_value": hodge_val,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": "SOS refutation size is zero"
        }

    upper_bound = math.sqrt(sos_size)

    return {
        "metric_name": "Hodge Index",
        "metric_value": hodge_val,
        "instances_tested": 1,
        "conjecture_holds": abs(hodge_val - upper_bound) <= 3,
        "counterexample": f"HodgeIndex({hodge_val}) > sqrt({sos_size})"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 89))  # Default to first 30 primes if

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(result
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"HodgeIndex({results[0]['metric_value']}) > sqrt
    else:
        print(f"RESULT: INCONCLUSIVE reason=unknown")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
etric_name': 'Hodge Index', 'metric_value': 28, 'instances_tested': 1, 'conjecture_holds': False, 'co
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 325, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 276, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 45, 'instances_tested': 1, 'conjecture_holds':
```

```
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 741, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 136, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 666, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 190, 'instances_tested': 1, 'conjecture_holds'
TRIAL: {'metric_name': 'Hodge Index', 'metric_value': 153, 'instances_tested': 1, 'conjecture_holds'

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c3d06598.py", line 86, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c3d06598.py", line 86, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                ~^^^^^^^^
KeyError: 'seed'
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> Safety rail: critic_challenge_falsified | original: The test results show that for multiple instances, the Hodge index of the quasi-Clifford algebra exceeds the square root of its smallest SOS refutatio | next: Investigate further to find a counterexample where the Hodge index is significantly higher than the square root of the SOS refutation size, or refine the conjecture to account for such exceptions.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15113
2	preregistration	ollama_remote	glm4:latest	0	0	10793
3	novelty	ollama_remote	glm4:latest	0	0	8535
4	novelty	ollama_remote	glm4:latest	0	0	9384
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14964
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11551
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10492
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9194
9	judge	ollama_remote	glm4:latest	0	0	12660

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 102687 ms total latency. Provider mix: {‘ollama_remote’: 9}

(full prompt+response transcripts available in `research/audit/d8441229b8b9.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/d8441229b8b9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/d8441229b8b9.tar.gz` (if generated)